

BATCH No:MI2090

DESIGN AND DEVELOPMENT OF A SMART LORA-BASED PIPELINE CORROSION HOTSPOT DETECTOR

*Minor Project-II report submitted in partial fulfillment of the requirement for award
of the degree of*

Bachelor of Technology in Computer Science & Engineering

By

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**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

May 2026

IOT2526426-Design & development of a smart LoRa-based Pipeline Corrosion Hotspot Detector

Project Objective :

The objective of the IoT-Based Pipeline Corrosion Monitoring and Hotspot Detection System is to develop an intelligent and automated monitoring solution that enhances pipeline safety and reliability. The system continuously monitors key environmental and corrosion-related parameters such as temperature, humidity, moisture levels, and electrochemical corrosion activity using integrated sensors. By analyzing these parameters in real time, the embedded Arduino-based controller identifies potential corrosion hotspots and evaluates pipeline conditions as SAFE or HOTSPOT. When predefined threshold values are exceeded, the system automatically activates warning mechanisms such as LEDs and buzzers while transmitting data through LoRa communication to a centralized web dashboard. This enables remote monitoring, real-time visualization, and timely maintenance decisions, thereby reducing manual inspection efforts, minimizing the risk of pipeline failure, and improving the overall efficiency of pipeline asset management.

System Implementation :

The implementation of the IoT-Based Pipeline Corrosion Monitoring and Hotspot Detection System Using LoRa Communication is divided into three major phases:

1. Data Acquisition:

The Arduino UNO continuously collects data from the DHT11 sensor, moisture sensor, and corrosion probe. The DHT11 measures temperature and humidity, the moisture sensor detects moisture levels around the pipeline, and the corrosion probe senses electrochemical changes that indicate corrosion activity. These sensor readings are periodically acquired and prepared for analysis.

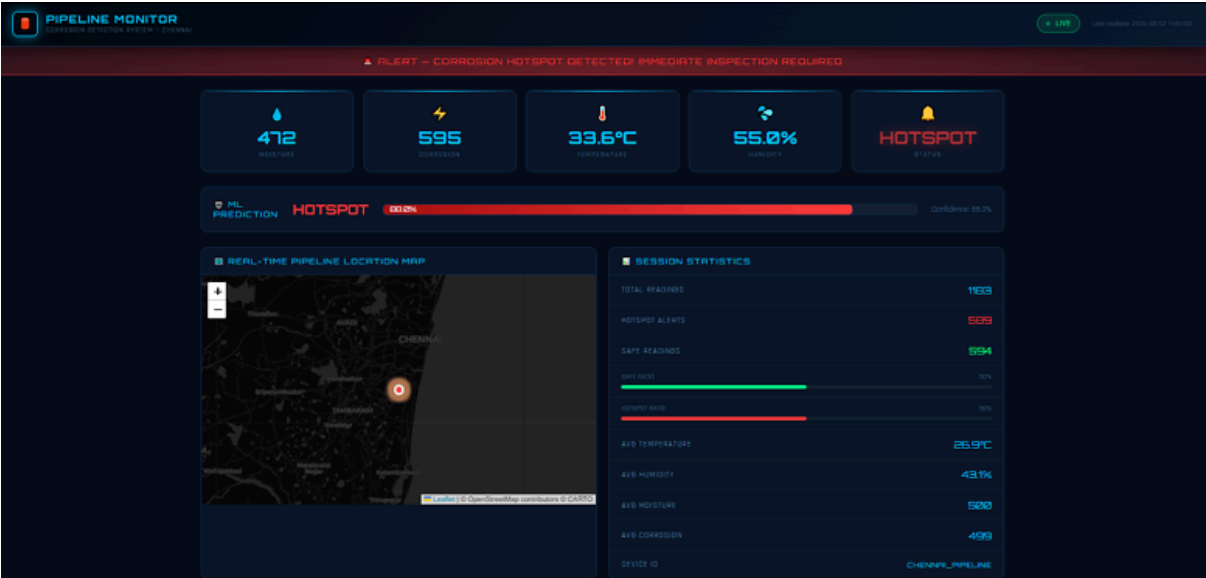
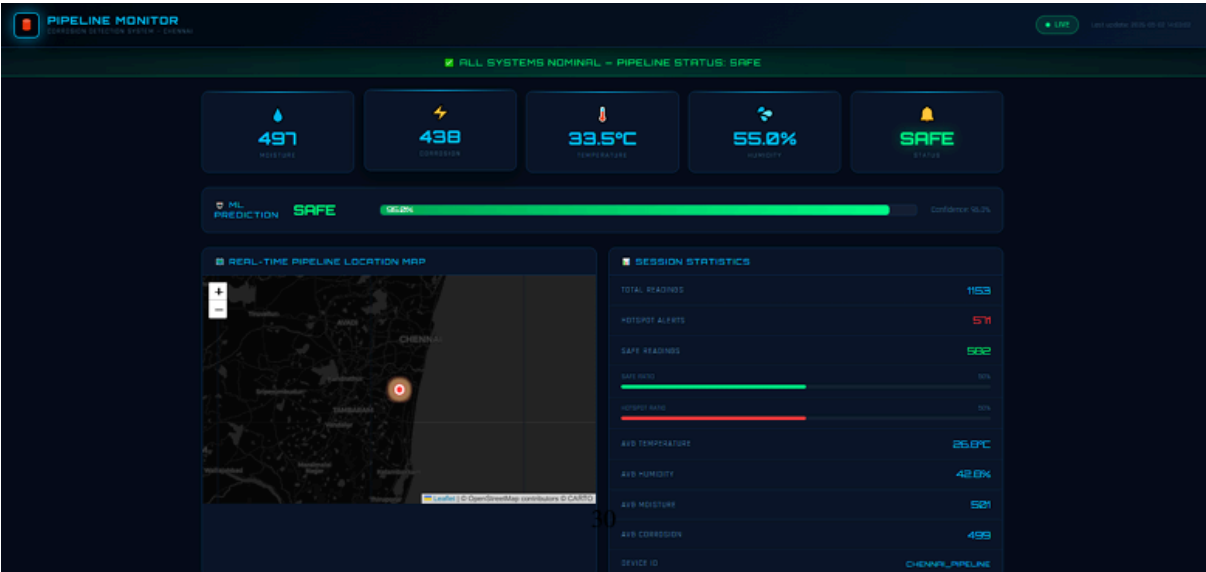
2. Corrosion Detection & Alert Mechanism:

The collected data is processed using predefined threshold values to evaluate the pipeline condition. Based on the corrosion level and environmental parameters, the system classifies the pipeline status as SAFE or HOTSPOT. When the measured values exceed the specified threshold, the system automatically activates the red LED and buzzer to provide an immediate warning. Under normal conditions, the green LED remains active, indicating safe operation.

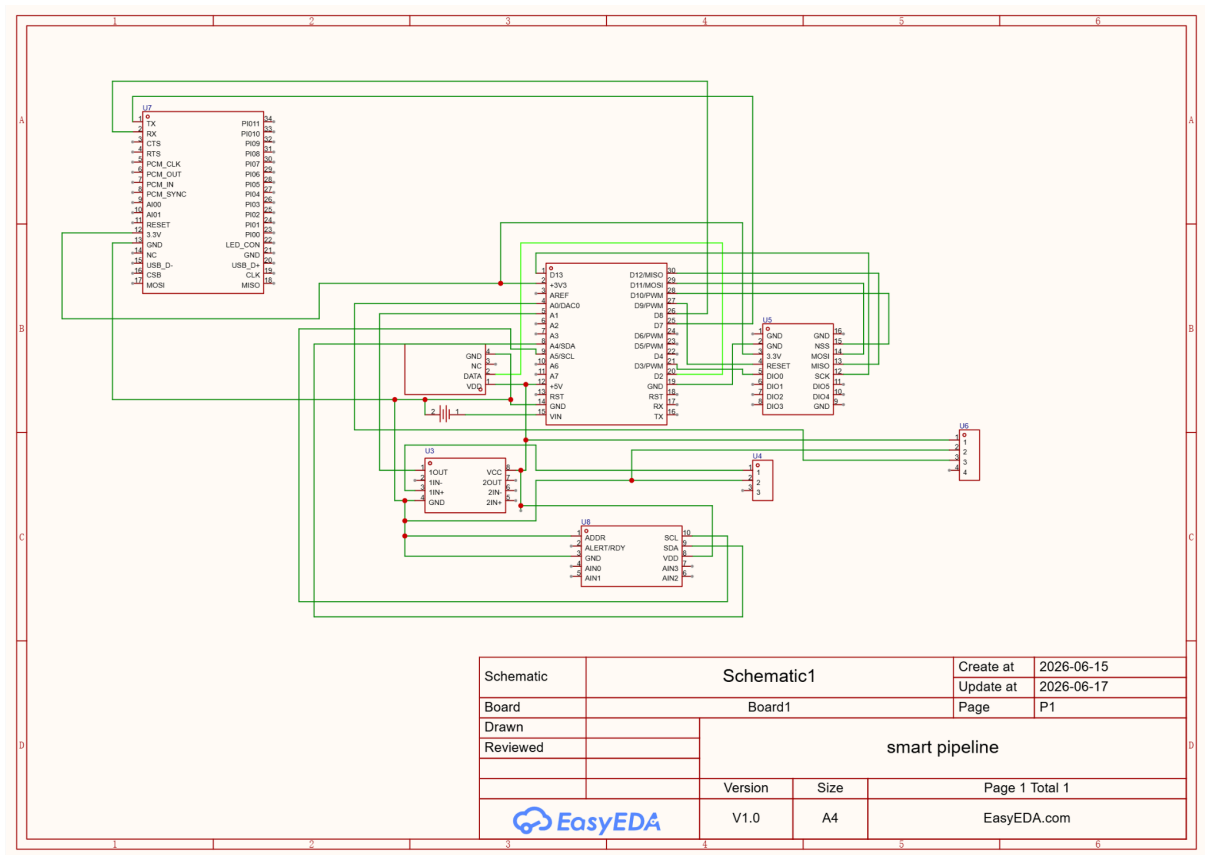
3. Data Transmission & Monitoring:

The processed data is transmitted through the LoRa (Ra-02) module for long-range wireless communication. The received data is stored in a database and displayed on a web-based dashboard. The dashboard provides real-time visualization of sensor readings, corrosion status, hotspot alerts, and historical records, enabling users to remotely monitor pipeline conditions and take timely maintenance actions.

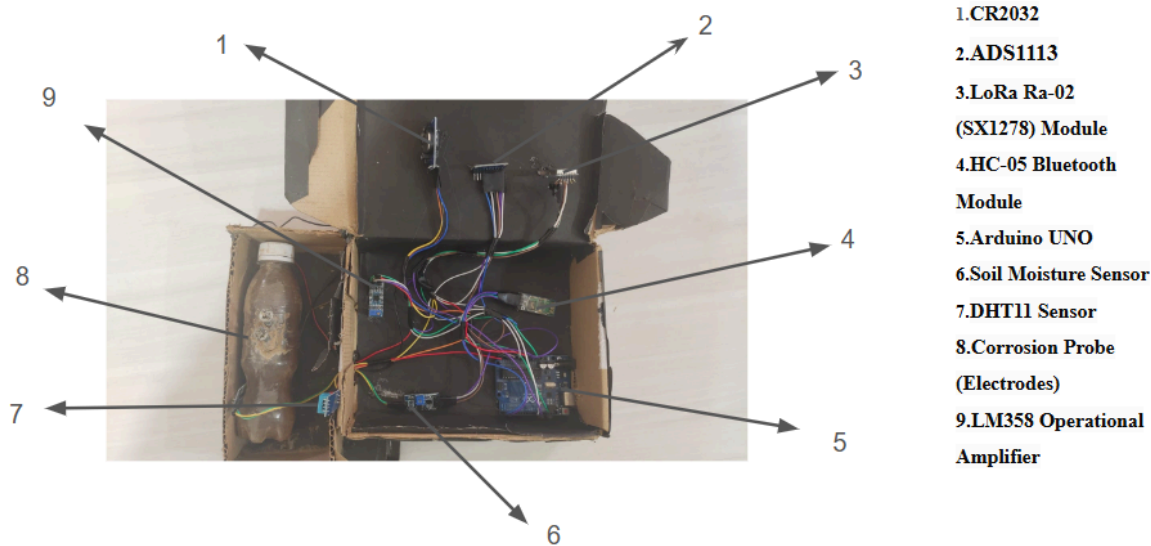
System Dashboard :



Circuit Diagram :



Prototype:(Hardware Components & Purpose)



Here is the practical breakdown of your physical architecture, mapped directly to the numbered components in your diagram:

1. CR2032 Battery

- **What it is:** A 3V lithium coin-cell battery used for low-power electronic circuits.
- **Why we use it:** It can provide backup power for low-power devices such as RTC modules and memory circuits.
- **How we use it:** In this project, it may be used as a backup power source for timekeeping or auxiliary circuits, but it is not used as the main power source for the Arduino UNO.

2. ADS1113

- **What it is:** A 16-bit Analog-to-Digital Converter (ADC) with I²C communication.
- **Why we use it:** It provides higher measurement accuracy than the built-in Arduino ADC, making it suitable for detecting small corrosion signal variations.
- **How we use it:** It converts the amplified analog signal from the LM358 into digital data and sends it to the Arduino UNO for processing.

3. LoRa Ra-02 (SX1278) Module

- **What it is:** A long-range wireless communication module based on LoRa technology.
- **Why we use it:** It enables low-power, long-distance data transmission between the monitoring unit and the remote monitoring station.
- **How we use it:** It transmits sensor readings and corrosion status information from the Arduino UNO to the web monitoring system.

4. HC-05 Bluetooth Module

- **What it is:** A wireless Bluetooth communication module.
- **Why we use it:** It allows short-range wireless communication between the monitoring system and nearby mobile devices.
- **How we use it:** It can be used to transmit sensor readings and system status information to a smartphone or computer for local monitoring.

5. Arduino UNO

- **What it is:** A microcontroller development board based on the ATmega328P.
- **Why we use it:** It acts as the main controller that processes sensor data and controls system operations.
- **How we use it:** It reads data from the sensors, evaluates corrosion conditions, generates alerts, and manages communication through LoRa and Bluetooth modules.

6. Soil Moisture Sensor

- **What it is:** A sensor used to measure the moisture content in the surrounding environment.
- **Why we use it:** Moisture is one of the major factors that accelerates pipeline corrosion.
- **How we use it:** The sensor continuously measures moisture levels near the pipeline and sends the readings to the Arduino UNO for analysis.

7. DHT11 Sensor

- **What it is:** A digital sensor that measures temperature and humidity.
- **Why we use it:** Environmental temperature and humidity significantly affect corrosion rates.
- **How we use it:** It continuously monitors ambient conditions and provides temperature and humidity data to the Arduino UNO.

8. Corrosion Probe (Electrodes)

- **What it is:** An electrochemical sensing probe used to detect corrosion activity on metal surfaces.
- **Why we use it:** It directly measures corrosion-related changes occurring on the pipeline.
- **How we use it:** The probe generates a small electrical signal corresponding to corrosion activity, which is then processed by the LM358 amplifier and ADS1113 ADC.

9. LM358 Operational Amplifier

- **What it is:** A dual operational amplifier used for analog signal conditioning.
- **Why we use it:** The signal generated by the corrosion probe is very weak and requires amplification before processing.
- **How we use it:** It amplifies and stabilizes the corrosion probe signal before sending it to the ADS1113 for accurate digital conversion.